Practical 9

Determination of insulation resistance value of motor windings

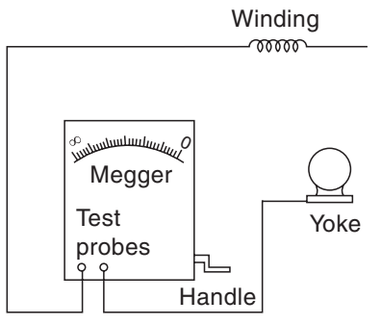
AIM

To determine the insulation value of a resistance of a given motor winding by using megger.

Instruments and apparatus required

S.no	Apparatus	Quantity
1	Given motor	1
2	Megger (1000 V)	1
3	Cutting plier	1
4	Screw driver	1
5	Double end spanner set	1

Connection diagram



Procedure

i. Megger is an instrument used to measure high value of resistance and it can also be used to measure the insulation value of the resistance of the motor windings.

ii. To find the value of insulation resistance, one terminal of the megger is connected with anyone terminal of winding and another terminal of motor is connected with the body of the motor.

iii. Rotate the handle of the megger about 160 revolutions per minute.

iv. Now the pointer of the megger moves gradually and shows the reading. That is the insulation value of the resistance of the winding.

v The reading should not be less than 1 mega ohm.

Measurement of insulation resistance value in DC machines

The windings in a DC machine are

1. Field winding

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